Power BI Dashboard Playbook

Bluestock Mutual Fund Analytics Capstone (Day 5), build in one day

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0. What you are building

A 4-page interactive dashboard, plus 2 hidden helper pages (drill-through and tooltip). Every page has at least 2 slicers. You will save four files in `dashboard/`:

File	How
`bluestock_mf_dashboard.pbix`	File > Save
`Dashboard.pdf`	File > Export > Export to PDF
`page1_industry.png` to `page4_sip_market.png`	Snipping Tool (`Win + Shift + S`)

Before you start (10 min):

- ☐ Power BI Desktop installed and open
- ☐ `bluestock_dashboard_data.xlsx` and `bluestock_theme.json` from the `dashboard/` folder of the project
- ☐ A Bluestock logo image (any PNG; use the company logo from the workspace site if you have it)
- ☐ File > Options > **Current File > Data Load > Time intelligence: turn OFF "Auto date/time"** (stops hidden date tables cluttering the model)
- ☐ File > Options > **Security > Map and filled map visuals: tick "Use Map and Filled map visuals"** (needed for Page 3; if this option is greyed out, use the bar-chart fallback described there)

1. Time plan (about 6 hours)

Time	Task
0:00 - 0:30	Load data, fix types, Calendar table
0:30 - 1:00	Relationships and measures
1:00 - 1:20	Theme, header band, page layout template
1:20 - 2:20	Page 1: Industry Overview
2:20 - 3:50	Page 2: Fund Performance (plus drill-through and tooltip)
3:50 - 4:50	Page 3: Investor Analytics
4:50 - 5:40	Page 4: SIP and Market Trends
5:40 - 6:10	Polish, check against section 8, export

Build **one page at a time and save after each**. Press `Ctrl + S` often.

2. Data model

2.1 Load

1. **Home > Get data > Excel workbook** > pick `bluestock_dashboard_data.xlsx`.
2. Tick **Select multiple items**, tick all 12 sheets, click **Transform Data**.
3. In Power Query, click each table in the left list and check:
   - `amfi_code` has the **ABC** (Text) icon in every table that has it
   - `date`, `month`, `transaction_date`, `launch_date`, `portfolio_date` have the **calendar** (Date) icon
4. **Home > Close & Apply**.

2.2 Calendar table

Modeling > New table, paste:

```
Calendar = ADDCOLUMNS(
    CALENDAR(DATE(2022,1,1), DATE(2026,5,31)),
    "Year", YEAR([Date]),
    "Month No", MONTH([Date]),
    "YearMonth", YEAR([Date]) * 100 + MONTH([Date]),
    "Month", FORMAT([Date], "MMM yyyy"))
```

Click the `Month` column > **Column tools > Sort by column > YearMonth** (not Month No, which would mix the same month of different years). Click the `Calendar` table > **Table tools > Mark as date table** > choose `Date`.

2.3 Relationships

Open **Model view** (third icon on the left). Drag the field from the 1 side onto the field in the many side. Each line must be **one-to-many (1 : *), single **direction,** active**.

1 side	Many side
`fund_master[amfi_code]`	`fund_scorecard[amfi_code]`
`fund_master[amfi_code]`	`scheme_performance[amfi_code]`
`fund_master[amfi_code]`	`nav_vs_benchmark[amfi_code]`
`fund_master[amfi_code]`	`transactions_enriched[amfi_code]`
`fund_master[amfi_code]`	`portfolio_holdings[amfi_code]`
`Calendar[Date]`	`aum_by_fund_house[date]`
`Calendar[Date]`	`monthly_sip_inflows[month]`
`Calendar[Date]`	`sip_vs_nifty_monthly[month]`
`Calendar[Date]`	`category_inflows[month]`
`Calendar[Date]`	`industry_folio_count[month]`
`Calendar[Date]`	`transactions_enriched[transaction_date]`
`Calendar[Date]`	`nav_vs_benchmark[date]`

`scheme_performance` and `fund_scorecard` also contain fund names. Always build slicers from **`fund_master`** so one slicer filters all fund tables.

Leave `benchmark_indices` unrelated (it is only used through `nav_vs_benchmark`).

2.4 Measures

Right-click the table name > **New measure** (create them in `fund_scorecard`, then move them to a table named `Measures` if you like: Home > Enter data > name it `Measures` > Load, then use the **Measure tools > Home table** drop-down). Paste one at a time.

```
Latest AUM (lakh crore) =
CALCULATE(SUM(aum_by_fund_house[aum_lakh_crore]),
          aum_by_fund_house[date] = MAX(aum_by_fund_house[date]))

Latest SIP Inflow (crore) =
CALCULATE(SUM(monthly_sip_inflows[sip_inflow_crore]),
          monthly_sip_inflows[month] = MAX(monthly_sip_inflows[month]))

Latest Folios (crore) =
CALCULATE(SUM(industry_folio_count[total_folios_crore]),
          industry_folio_count[month] = MAX(industry_folio_count[month]))

Schemes (10 AMCs) =
CALCULATE(SUM(aum_by_fund_house[num_schemes]),
          aum_by_fund_house[date] = MAX(aum_by_fund_house[date]))

Active SIP Accounts (crore) =
CALCULATE(SUM(monthly_sip_inflows[active_sip_accounts_crore]),
          monthly_sip_inflows[month] = MAX(monthly_sip_inflows[month]))

SIP Accounts YoY % =
VAR latest = MAX(monthly_sip_inflows[month])
VAR cur  = CALCULATE(SUM(monthly_sip_inflows[active_sip_accounts_crore]),
monthly_sip_inflows[month] = latest)
VAR prev = CALCULATE(SUM(monthly_sip_inflows[active_sip_accounts_crore]),
monthly_sip_inflows[month] = EDATE(latest, -12))
RETURN DIVIDE(cur - prev, prev)

Total Invested (crore) =
DIVIDE(CALCULATE(SUM(transactions_enriched[amount_inr]),
                 transactions_enriched[transaction_type] <> "Redemption"),
10000000)

Net Flow (crore) =
DIVIDE(SUM(transactions_enriched[signed_amount_inr]), 10000000)

Avg SIP Amount =
CALCULATE(AVERAGE(transactions_enriched[amount_inr]),
          transactions_enriched[transaction_type] = "SIP")

Investors = DISTINCTCOUNT(transactions_enriched[investor_id])

Top Performer =
MAXX(TOPN(1, fund_scorecard, fund_scorecard[score]),
fund_scorecard[scheme_name])
```

Set the format (click the measure > **Measure tools** ribbon):

Measure	Format	Decimals
Latest AUM (lakh crore)	Decimal number	2
Latest SIP Inflow (crore)	Whole number, thousands separator on	0
Latest Folios (crore), Active SIP Accounts (crore)	Decimal number	2
Schemes (10 AMCs), Investors	Whole number, thousands separator on	0
SIP Accounts YoY %	Percentage	1
Total Invested (crore), Net Flow (crore)	Decimal number	0
Avg SIP Amount	Currency, Rs	0

Quick test: put each measure in a Card visual on a blank page and compare with section 8 before building real pages.

3. Page template (do once, then duplicate)

1. **View > Themes > Browse for themes** > `bluestock_theme.json`.
2. **Page size:** Format pane > Canvas settings > Type = 16:9 (1280 x 720). Page background `#F4F7FB`.
3. **Header band:** Insert > Shapes > Rectangle. Position X 0, Y 0, Width 1280, Height 56. Fill `#1F4E79`, no outline.
4. **Page title:** Insert > Text box on the band, X 20, Y 10, W 600, H 36. Font Segoe UI Semibold 20, white. Text changes per page, e.g. "Industry Overview".
5. **Logo:** Insert > Image, X 1190, Y 8, W 70, H 40.
6. **Page navigator:** Insert > Buttons > Navigator > Page navigator. X 640, Y 10, W 530, H 36. Format: text white, selected state fill `#C55A11`.
7. **Footer note:** text box X 20, Y 695, W 1240, H 20, 8 pt grey: "Source: AMFI India, mfapi.in, NSE/BSE public data; investor transactions are simulated. Educational use only."

Set exact positions in **Format > General > Properties** (X position, Y position, Width, Height). **Duplicate the page 3 times** (right-click the page tab > Duplicate page) and rename them `1 Industry Overview`, `2 Fund Performance`, `3 Investor Analytics`, `4 SIP and Market Trends`.

Slicer style (all slicers): select the slicer > Format > Slicer settings > Options > Style = **Dropdown**. This saves space and keeps the layout clean.

4. Page 1: Industry Overview

Layout (X, Y, Width, Height in pixels):

Visual	Position	Fields and settings
Slicer: Fund house	20, 66, 300, 44	`aum_by_fund_house[fund_house]`, dropdown, multi-select
Slicer: Year	340, 66, 200, 44	`Calendar[Year]`, dropdown
Card 1	20, 120, 295, 96	`Latest AUM (lakh crore)`; title "Top-10 AMC AUM (lakh crore)"
Card 2	335, 120, 295, 96	`Latest SIP Inflow (crore)`; title "SIP inflow (Rs crore)"
Card 3	650, 120, 295, 96	`Latest Folios (crore)`; title "Folios (crore)"
Card 4	965, 120, 295, 96	`Schemes (10 AMCs)`; title "Schemes

Visual	Position	Fields and settings
		(top 10 AMCs)"
Line chart	20, 230, 620, 400	X: `aum_by_fund_house[date]` (choose the plain date, not the Date Hierarchy); Y: Sum of `aum_lakh_crore`; title "AUM of the top 10 fund houses"
Bar chart	660, 230, 600, 400	Y: `fund_house`; X: `Latest AUM (lakh crore)`; sort descending; data labels on; title "AUM by fund house, latest report"
Text box	20, 640, 1240, 48	"Industry context (AMFI, Dec 2025): AUM Rs 81 lakh crore, 1,908 schemes, 26.12 crore folios."

Notes

- The brief shows "Rs 81 lakh crore" and "1,908 schemes" as industry totals. These are **not in the data** (the data covers 10 fund houses), so show them only in the text box and label the cards "Top-10 AMC".
- Line chart: with only 9 reporting dates the line is short. Turn on **Markers** (Format > Visual > Markers).
- Add the callout to the bar chart: Format > Data labels > Display units = None.

5. Page 2: Fund Performance (plus 2 helper pages)

Visual	Position	Fields and settings
Slicer: Fund house	20, 66, 230, 44	`fund_master[fund_house]`
Slicer: Category	265, 66, 200, 44	`fund_master[category]`
Slicer: Plan	480, 66, 180, 44	`fund_master[plan]`
Slicer: Scheme	675, 66, 585, 44	`fund_master[scheme_name]`; Format > Slicer settings > Selection > turn **Single select ON**; select "SBI Bluechip Fund - Regular Plan - Growth"
Scatter chart	20, 120, 600, 300	X: Average of `scheme_performance[return_3yr_pct]`; Y: Average of `scheme_performance[std_dev_ann_pct]`; Size: Sum of `scheme_performance[aum_crore]`; Legend: `fund_master[category]`; Details: `scheme_performance[scheme_name]`; title "3-year return vs risk (bubble = AUM)"
Table	640, 120, 620, 300	`fund_scorecard`: `rank_overall`, `scheme_name`, `score`, `cagr_3yr`, `sharpe`, `alpha`, `max_drawdown`, `tier`
Line chart	20, 435, 1240, 255	X: `nav_vs_benchmark[date]`; Y: Average of `nav_rebased` and Average of `benchmark_rebased`; title "NAV vs benchmark (start = 100)"; legend on

Scatter chart setup

- Axis titles: "3-year return (%)" and "Volatility (% a year)".

- Legend colours: Equity `#1F4E79`, Debt `#C55A11`.

Table formatting

- Format > Cell elements > **Data bars** on `score`, **Background colour (colour scale)** on `sharpe` (red `#C0392B` to green `#2E8B57`).
- Show `cagr_3yr` and `max_drawdown` as percentages: select the column > Column tools > Format = Percentage, 1 decimal.
- Click a column header to sort. Sort by `rank_overall` ascending at the start.
- `alpha` here comes from a regression on Nifty 100 and is not meaningful in this dataset (see the data notes in the README). Keep it because the brief's scorecard uses it.

NAV vs benchmark chart: it follows the Scheme slicer. If the lines look flat or odd, the Scheme slicer is not set to a single fund. To show a fund by clicking a table row instead, clear the Scheme slicer selection; clicking a table row will then filter the line chart.

Cross-filter control: select the Fund house slicer > Format > **Edit interactions**. Make sure it filters the scatter, table and line chart (the "filter" icon on each).

5.1 Drill-through page "Fund Detail"

1. Add a new page, name it `Fund Detail`.
2. In the **Filters pane > Drill through**, drag `fund_master[scheme_name]` into "Add drill-through fields here". Keep "Keep all filters" ON.
3. Add: Card `Top Performer` is not needed here. Add cards for `fund_scorecard[score]`, `fund_scorecard[sharpe]`, `fund_scorecard[max_drawdown]`; the NAV vs benchmark line chart; a column chart of `portfolio_holdings` (`sector` and Sum of `weight_pct`).
4. Power BI adds a back button automatically.
5. Test: on Page 2 right-click a fund in the table > **Drill through > Fund Detail**.
6. Right-click the `Fund Detail` tab > **Hide page**.

5.2 Tooltip page "Fund Tooltip"

1. Add a new page, name it `Fund Tooltip`. In Format > Page information turn **Allow use as tooltip ON**. In Canvas settings choose type **Tooltip**.
2. Add 3 cards: `fund_scorecard[sharpe]`, `fund_scorecard[cagr_3yr]`, `fund_scorecard[max_drawdown]`.
3. On Page 2 select the scatter chart > Format > **General > Tooltips** > Type = Report page > Page = `Fund Tooltip`.
4. Hide the tooltip page.

6. Page 3: Investor Analytics

Visual	Position	Fields and settings
Slicer: State	20, 66, 300, 44	`transactions_enriched[state]`
Slicer: Age group	335, 66, 300, 44	`transactions_enriched[age_group]`
Slicer: City tier	650, 66, 250, 44	`transactions_enriched[city_tier]`
Card 1	20, 120, 295, 80	`Total Invested (crore)`
Card 2	335, 120, 295, 80	`Net Flow (crore)`

Visual	Position	Fields and settings
Card 3	650, 120, 295, 80	`Avg SIP Amount`
Card 4	965, 120, 295, 80	`Investors`
Map	20, 215, 620, 475	**Filled map**; Location: `state` (set the field's Data category to **State or Province**: Column tools > Data category); Colour saturation: `Total Invested (crore)`
Donut	660, 215, 300, 230	Legend: `transaction_type`; Values: Sum of `amount_inr`; show percentages
Bar chart	975, 215, 285, 230	X: `age_group`; Y: `Avg SIP Amount`; sort by age group ascending
Line chart	660, 460, 600, 230	X: `transactions_enriched[month]` (plain date); Y: `Total Invested (crore)` and `Net Flow (crore)`

If the map does not show states (blank or "map visuals disabled"): replace it with a horizontal **bar chart**: Y `state`, X `Total Invested (crore)`, sorted descending. The data has 12 states, so it reads well.

Tip: the donut shows share of **amount**: Lumpsum is far larger than SIP by value, even though SIPs are the most frequent type. Add a second donut or a tooltip with `Count of transaction_type` if you want to show the transaction-count view as well.

7. Page 4: SIP and Market Trends

Visual	Position	Fields and settings
Slicer: Year	20, 66, 250, 44	`Calendar[Year]`
Slicer: Category	285, 66, 300, 44	`category_inflows[category]`
Card 1	20, 120, 295, 80	`Latest SIP Inflow (crore)`
Card 2	335, 120, 295, 80	`Active SIP Accounts (crore)`
Card 3	650, 120, 295, 80	`SIP Accounts YoY %`
Combo chart	20, 215, 760, 230	Visual: **Line and clustered column chart**. X: `sip_vs_nifty_monthly[month]`; Column Y: Sum of `sip_inflow_crore`; Line Y (secondary axis): Sum of `nifty50_month_end`. Title "Monthly SIP inflow (bars) vs Nifty 50 (line)"
Heat map	20, 455, 760, 235	**Matrix**. Rows: `category_inflows[category]`; Columns: `category_inflows[month]` (plain date, format `MMM yy`); Values: Sum of `net_inflow_crore`. Format > Cell elements > Background colour > rules or gradient: minimum `#C0392B`, centre 0 `#FFFFFF`, maximum `#2E8B57`
Bar chart	800, 215, 460, 230	Y: `category`; X: Sum of `net_inflow_crore`; **Filters pane > Top N**: Top 5 by `net_inflow_crore`; title "Top 5 categories by net inflow, FY 2024-25"
Line chart	800, 455, 460, 235	X: `industry_folio_count[month]`; Y: Sum of `total_folios_crore`; title

Visual	Position	Fields and settings
		"Industry folios (crore)"

Notes

- Combo chart: Format > X axis > Type = **Categorical** so each month shows. Rename the secondary axis "Nifty 50".
- `category_inflows` covers Apr 2024 to Mar 2025 only. The Year slicer filters that matrix to 2024 or 2025; this is expected.
- Mark the all-time high: select the column for Dec 2025 > Format > Data colors > Show all > set Dec 2025 to `#C0392B`, and add a text box note "Rs 31,002 crore, all-time high".

8. Numbers to check (your dashboard should match)

Where	Expected value
Page 1, Top-10 AMC AUM	**62.74** lakh crore (as of 31 Dec 2025)
Page 1, SIP inflow	**31,002** crore
Page 1, Folios	**26.12** crore
Page 1, Schemes	**1,522**
Page 1, biggest bars	SBI 12.50, ICICI Prudential 10.74, HDFC 9.30 (lakh crore)
Page 2, top table row	**Mirae Asset Large Cap Fund - Regular - Growth**, score **84.8**
Page 3, Total Invested	about **228** crore
Page 3, Net Flow	about **103** crore
Page 3, Avg SIP Amount	about **Rs 11,018**
Page 3, Investors	**5,000**
Page 3, donut (by amount)	Lumpsum **58.5%**, Redemption **35.3%**, SIP **6.2%**
Page 3, avg SIP by age	18-25: 10,953; 26-35: 10,987; 36-45: 10,886; 46-55: 11,137; 56+: **11,575** (highest)
Page 3, map or bars	**12** states
Page 4, Active SIP accounts	**9.35** crore
Page 4, SIP Accounts YoY	**+10.0%** (8.50 to 9.35 crore)
Page 4, Top 5 categories (net inflow, crore)	Liquid 4,51,275; Sectoral/Thematic 1,03,829; Flexi Cap 63,989; Large & Mid Cap 57,752; Short Duration 55,530

If a number is off, the usual cause is a missing relationship or a slicer left on a selection. Clear all slicers and check again.

9. Polish checklist

- ☐ Every page has the header band, title, logo, page navigator and footer
- ☐ Every page has **2 or more slicers**
- ☐ Every visual has a title; no default "Sum of ..." titles
- ☐ Fonts consistent (Segoe UI); no overlapping visuals (View > Gridlines, Snap to grid)
- ☐ Axis titles on scatter and combo charts; units in every title (crore, lakh crore, %)
- ☐ Numbers formatted: no 10-digit decimals
- ☐ Drill-through works; tooltip appears when hovering over a scatter bubble
- ☐ Hidden pages: `Fund Detail`, `Fund Tooltip`
- ☐ Alt text on the main visuals (Format > General > Alt text)

- ☐ File > Save As `bluestock_mf_dashboard.pbix`
- ☐ Open each page with no slicers selected before taking screenshots

10. Export

1. **PDF:** File > Export > **Export to PDF**. Save as `Dashboard.pdf`. Check that all 4 pages are in it.
2. **Screenshots:** open each page, clear slicers, `Win + Shift + S`, drag over the whole canvas. Save as `page1_industry.png`, `page2_performance.png`, `page3_investor.png`, `page4_sip_market.png`.
3. **Optional publish:** Home > Publish (needs a work or school account). If you cannot sign in, skip it and say so in the report.
4. Copy the 6 files into the project's `dashboard/` folder, then `git add dashboard` and commit.

11. Demo video script (about 2 minutes)

Record with Xbox Game Bar (`Win + G`) or Zoom. Say roughly:

1. (10 s) "This is the Bluestock Mutual Fund Analytics dashboard. It has four pages built from AMFI public data."
2. (25 s) **Page 1:** "The top 10 fund houses manage 62.7 lakh crore. SBI is the largest at 12.5. SIP inflows reached 31,002 crore in December 2025 and folios 26.12 crore." Use the fund house slicer.
3. (35 s) **Page 2:** "Each bubble is a scheme: return against risk. The scorecard ranks funds on return, Sharpe, alpha, cost and drawdown." Click a fund, show the NAV vs benchmark line, drill through to Fund Detail.
4. (30 s) **Page 3:** "Investors by state, transaction type and age group. Lumpsum is 58% of the amount but SIP is the most frequent." Change the age slicer.
5. (15 s) **Page 4:** "SIP inflows rise over time. The heat map shows liquid funds took the largest net inflow in FY25."
6. (5 s) "All code and data are in the GitHub repo. Thank you."

12. Troubleshooting

Problem	Fix
"External table is not in the expected format"	You used a `.xls` file. Load `bluestock_dashboard_data.xlsx` with the Excel connector, or CSVs with Text/CSV
Numbers do not change when a slicer is used	Missing relationship. Check Model view; every table in section 2.3 must have a line
A relationship cannot be created ("many-to-many")	The key has duplicates or different types. Both `amfi_code` columns must be Text; `fund_master[amfi_code]` must be unique (40 rows)
Visual shows "(Blank)"	The field is from an unrelated table. Use fields from related tables only
Date axis shows Year > Quarter > Month	Click the arrow next to the date field in the well and choose the plain **date** name, not "Date Hierarchy"
Matrix shows months out of order	Use the real date column (`month`), not text
Map is empty	Enable map visuals (section 0), set Data category = State or Province; otherwise use the bar-chart fallback
NAV vs benchmark line is wrong	Make sure the Scheme slicer is single-select with one fund chosen

Problem	Fix
Cards show a huge number	A card is summing every date. Use the "Latest …" measures, not raw columns
Slow performance	Close other apps; `nav_vs_benchmark` has 46,000 rows, which is fine, but avoid putting all 40 funds on one line chart
A measure shows an error	Check the table and column names match exactly (copy from section 2.4)

When anything looks wrong, send a screenshot and the visual's field wells, and I can tell you what to change.